

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

ASSESSMENT TOOL 2–TECHNICAL PRESENTATION

Course Code	CSA0315	Course Title	Data Structures
CO Assessed	CO1	Assessment	Assessment Tool 2 : Technical presentation
Weightage	35% of CIA	Total Marks	100
CO1 Statement	<i>Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity. (BL2, BL3)</i>		
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Section/Batch	Slot D /2025	Date	06-10-2026

Code of Conduct:

I, Abarajithan S (192521299), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Abarajithan S

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood	
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach	
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals	
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides	
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness	



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CO1-Assessment Tool 2

Implementation of LRU Cache Using Data Structures

CSA0315-Data Structures

GUIDED BY:
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Introduction

LRU Cache System :

- ✓ LRU = Least Recently Used
- ✓ It is a cache replacement technique.
- ✓ Stores recently used data.
- ✓ Removes the least recently used data.
- ✓ Helps to increase system speed.

Data Structure Used

- ✓ Hash Map + Doubly Linked List
- ✓ Hash Map → Finds data quickly.
- ✓ Doubly Linked List → Maintains usage order.
- ✓ Recently used data → Front
- ✓ Least recently used data → Rear
- ✓ Both provide fast operations.

How LRU Works

Steps :

- ✓ Check if data is in cache.
- ✓ If found → **Cache Hit**.
- ✓ Move the data to the front.
- ✓ If not found → **Cache Miss**.
- ✓ Add new data to the front.
- ✓ If cache is full → Remove the last item.

Performance

Why LRU is Fast

- Search $\rightarrow O(1)$
- Insert $\rightarrow O(1)$
- Delete $\rightarrow O(1)$
- Reduces memory access time.
- Improves application response time.

Applications & Conclusion

Applications:

- Web browsers
- Operating systems
- Databases
- CPU caching
- Mobile applications

Conclusion :

- LRU keeps useful data readily available.
- Hash Map + Doubly Linked List gives **fast access**.
- Therefore, LRU **improves system performance**.